

6.3.3 Electromagnetism

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) magnetic flux ϕ ; the unit weber; $\phi = BA\cos\theta$	
(b) magnetic flux linkage	
(c) Faraday's law of electromagnetic induction and Lenz's law	HSW2, 7
(d) (i) e.m.f. = – rate of change of magnetic flux linkage; $E = -\frac{\Delta(N\phi)}{\Delta t}$ (ii) techniques and procedures used to investigate magnetic flux using search coils	M3.9 HSW2, 8
(e) simple a.c. generator	HSW8
(f) (i) simple laminated iron-cored transformer; $\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}$ for an ideal transformer	M0.3 HSW9
(ii) techniques and procedures used to investigate transformers.	HSW3, 9